trained on randomized topic tags, implying that early in training the model has removed the context necessary to identify topic (below even the inadequate contextual information memorized by a model with random labels), which depends on the general vocabulary in a sentence rather than a local sequence. Over time correlation rises, possibly because the model permits more long-distance context to be encoded. Khandelwal et al. (2018) found that LSTMs remember content words like nouns for more time steps than they remember function words like prepositions and articles. We hypothesize that the LM's slower stabilization on topic is related to this phenomenon, since it must depend on content words, and its ability to remember them increases throughout training.

The encoder layer exhibits very different patterns. Because the representation produced by the encoder layer is local to the word, the nuances that determine how a word is tagged in context cannot be learned. The encoder layers are all highly similar to each other, which suggests that the unigram representations produced by the encoder are less dependent on the particular end task of the neural network. Similarity between the encoders declines over time as they become more specialized towards the language modeling task. This decline points to some simple patterns which are learned for all language tasks, but which are gradually replaced by representations more useful for language modeling. This process may even be considered a naturally occurring analog to the common practice of initializing the encoder layer as word embeddings pretrained an unrelated task such as skipgram or CBOW (Mikolov et al., 2013). It seems that the 'easy' word properties, which immediately improve performance, are similar regardless of the particular language task.

At h^1, the correlation shows a clear initial decline in similarity for all tasks. This seems to point to an initial representation that relies on simple shared properties, which in the first stage of training is gradually dissolved before the layer begins to converge on a structure shared with each tag predictor. It may also be linked to the information bottleneck learning phases explored by Shwartz-Ziv and Tishby (2017). They suggest that neural networks learn by first maximizing the mutual information between the input and internal representation, then minimizing the mutual information between the internal representation and output. The network thus initially learns to effectively represent the input, then compresses this representation, keeping only the elements relevant to the output.[3] If the LM begins by maximizing mutual information with input, because the input is identical for the LM and tag models it may lead to these similar initial representations, followed by a decline in similarity as the compression narrows to properties specific to each task.

4.2 SVCCA on Input Tagging

Our second conceptual shift is to focus on output tag prediction—asking what a representation encodes about the *next* output word, rather than what it has encoded about words it has already observed in the input. What effect does this have? Since we already studied output tags in the previous set of experiments, here we consider input tags, in the style of most diagnostic classifier analysis (Figure 9). The learning dynamics are similar to those for tag prediction, but the UDP POS tagger decreases dramatically in all correlations while the GMB-trained taggers[4] often increase slightly. While the shapes of the lines are similar, UDP POS no longer consistently dominates the other tasks in recurrent layer correlation. Instead, we find the more granular PTB POS tags lead to the most similar representations.

5 Discussion and Conclusions

We find clear patterns in the encoding of linguistic structure with SVCCA, in contrast to the weaker results from a less responsive diagnostic classifier. Because SVCCA proves so much more sensitive than the diagnostic classifiers currently in use, we believe that future work on measuring the encoding of linguistic structure should use the similarity of individual modules from independently trained tag predictors rather than the performance of tag predictors trained on a particular representation.

This system should also be of interest because it is efficient. To train a diagnostic classifier, we must run a forward pass of the LM for each forward pass of the auxiliary model, while SVCCA only requires the LM to run on the test set. A further efficiency gain is particular to studying learn-

[3] This memorization–compression learning pattern parallels the memorization–generalization of the first half of the U-shaped curve exhibited by human children learning irregular word forms. **?** observe similar patterns when artificially modeling inflection.

[4] PTB POS, SEM (fine), and SEM (coarse)

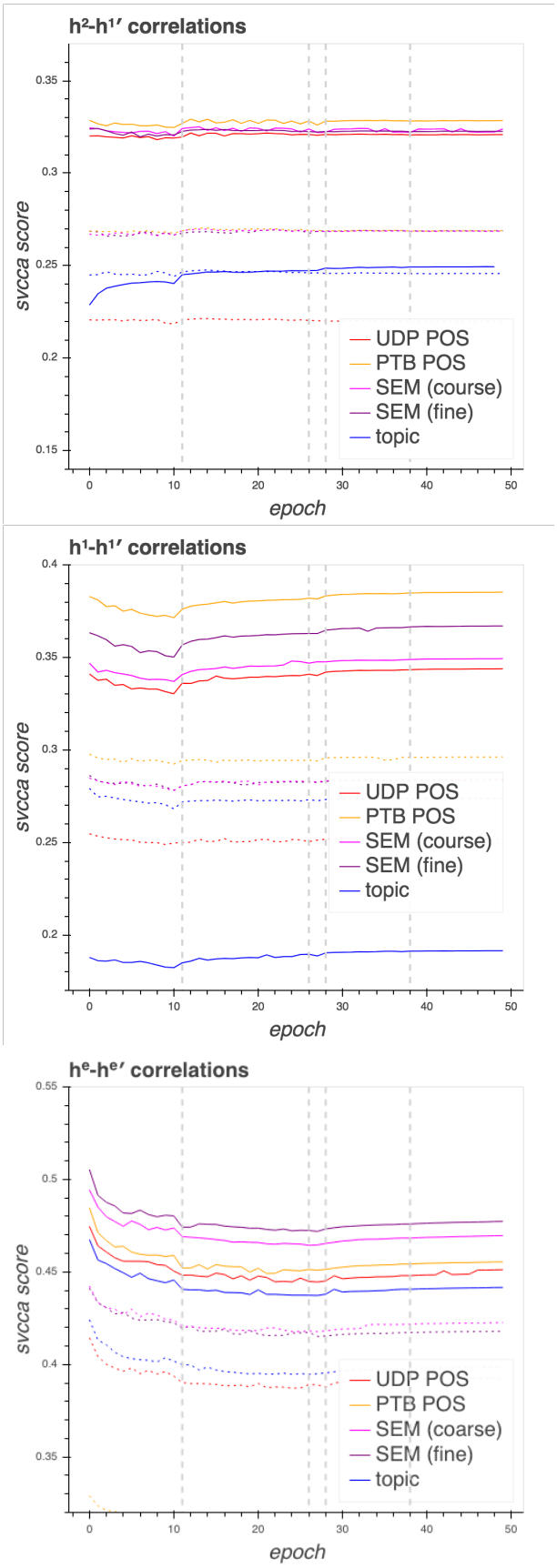

Figure 9: SVCCA correlation scores between LM activations when predicting x_{t+1} and tagger activations when labeling y_t. Dotted lines use shuffled tags. Gray vertical lines mark when the step size is rescaled.

ing dynamics: we train only one tagger and compare it to different versions of the LM over training, but for standard probing, we must train a new version of each layer's tagger at each epoch. Our SVCCA experiments in Figure 8 ran in hours, while the diagnostic classifier experiments in Figure 7 ran for days.

Our method holds another, more subtle advantage. Our analysis provides an alternative view of what it means for a model to *encode* some linguistic trait. The literature on analyzing neural networks includes a broad spectrum of interpretations about what it means to encode a property. At one end of the spectrum lies the purely informational view (e.g., mutual information; **?**). Mutual information is a very flexible view, but it requires us to compute theoretical information content, which in practice can only be estimated. Furthermore, information can be represented without being used, as shown by **?**, who found that NMT systems often predicted tense according to a diagnostic classifier but did not produce the correct tense as output. The other end of the spectrum is focused on the structure of the representation space (e.g., the features and the property in question are linearly similar; Alishahi et al., 2017). Analyzing structural similarity should remedy the shortcomings of the informational view, but most intermediate representations are not targeted to extract the property in question through a linear transformation, and failing to be interpretable through such simple extraction should not be equated to a failure to encode that property.

Most of the literature on analyzing representations, by probing with a more complex architecture, seeks the flexibility of mutual information with the concreteness and tractability of the structural view – but instead obscures the strict information view without offering interpretable information about the structure, because the architecture of a diagnostic classifier affects its performance. It should not be surprising that representational quality as measured by such systems is a poor indicator of translation quality (**?**). SVCCA, in contrast, is a structural view that does not directly compare an activation that targets word prediction with a particular tag, but instead compares that activation with one targeting the prediction of the tag.

Let us consider a specific common probing method. What do we learn about the LM when a

feedforward network cannot extract tag information directly from the embedding layer, but can from a recurrent layer? It may be tempting to conclude that tag information relies heavily on context, but consider some alternative explanations. If the embedding encodes the tag to be interpreted by a recurrent layer, a feedforward network may not be capable of representing the function to extract that tag because it does not have access to a context vector for aiding interpretation of the hidden layer. Perhaps its activation functions cover a different range of outputs. By directly comparing LSTM layers to LSTM layers and embedding layers to embedding layers, we respect the shape of their outputs and the role of each module within the network in our analysis.

The results of our analysis imply that early in training, representing part of speech is the natural way to get initial high performance. However, as training progresses, it increasingly benefits the model to represent categories with longer-range dependencies, such as topic.

6 Future Work

One direction for future work is exploring how generalization interacts with the correlations between LMs and tag predictors. It may be that a faithful encoding of a property like POS tag indicates that the LM is relying more on linguistic structure than on memorizing specific phrases, and therefore is associated with a more general model.

If these measurements of structure encoding are associated with more general models, we might introduce regularizers or other modifications that explicitly encourage correlation with a tagging task.

Combes et al. (2018) identified the phenomenon of *gradient starvation*, meaning that while frequent and unambiguous features are learned quickly in training, they slow down the learning of rarer features. For example, artificially brightening images according to their class leads to a delay in learning to represent the more complex natural class features. Although it is tempting to claim that semantic structure is learned using syntactic structure as natural scaffolding, it is possible that the simple predictive power of POS is acting as an attractor and starving semantic features that are rarer and more ambiguous. A possible direction for future work would be to explore which of these explanations is true, possibly by decorrelating particular aspects of linguistic structure from language modeling representations.

The techniques in this paper could be applied to better understand the high performance of a system like ELMo (Peters et al., 2018). Different layers in such a system are useful for different tasks, and this effect could be understood in terms of the gradual divergence between the layers and their respective convergence to representations geared toward a single task.

Acknowledgements

We thank Denis Emelin, Sameer Bansal, Toms Bergmanis, Maria Corkery, Sharon Goldwater, Sorcha Gilroy, Aibek Makazhanov, Yevgen Matusevych, Kate McCurdy, Janie Sinclair, Ida Szubert, Nikolay Bogoychev, Clara Vania, and the anonymous reviewers for helpful discussion and comments on drafts. We thank Matthew Summers for assistance with visualisations.

References

Afra Alishahi, Marie Barking, and Grzegorz Chrupała. 2017. Encoding of phonology in a recurrent neural model of grounded speech. In *Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017)*, pages 368–378.

Yonatan Belinkov, Nadir Durrani, Fahim Dalvi, Hassan Sajjad, and James Glass. 2017. What do Neural Machine Translation Models Learn about Morphology? In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 861–872, Vancouver, Canada. Association for Computational Linguistics.

Yonatan Belinkov, Lluís Màrquez, Hassan Sajjad, Nadir Durrani, Fahim Dalvi, and James R. Glass. 2018. Evaluating Layers of Representation in Neural Machine Translation on Part-of-Speech and Semantic Tagging Tasks. *CoRR*, abs/1801.07772.

Johannes Bjerva, Barbara Plank, and Johan Bos. 2016. Semantic Tagging with Deep Residual Networks. In *COLING*.

Terra Blevins, Omer Levy, and Luke Zettlemoyer. 2018. Deep RNNs Encode Soft Hierarchical Syntax. *arXiv:1805.04218 [cs]*. ArXiv: 1805.04218.

Johan Bos, Valerio Basile, Kilian Evang, Noortje J Venhuizen, and Johannes Bjerva. 2017. The groningen meaning bank. In *Handbook of linguistic annotation*, pages 463–496. Springer.

Remi Tachet des Combes, Mohammad Pezeshki, Samira Shabanian, Aaron Courville, and Yoshua Bengio. 2018. On the Learning Dynamics of Deep Neural Networks. *arXiv:1809.06848 [cs, stat]*. ArXiv: 1809.06848.